Adapter Scaling Trade-off and Timestep-Conditioned Scheduling in Multi-view Diffusion Texture Generation

Abstract: Multi-view diffusion models have become an important technical route for 3D texture generation, where adapter modules are commonly used to inject geometric conditions such as normals and depth maps to improve cross-view consistency and geometric alignment. However, the adapter scaling strength during inference is often selected empirically, and its influence on shape fidelity and texture detail remains insufficiently analyzed. This paper systematically studies adapter scaling in geometry-controlled multi-view texture generation. We find that a larger adapter scale can improve structural metrics such as FG-SSIM, but it may also suppress the texture synthesis capability of the base diffusion model, leading to weakened color variation, flattened surfaces, and loss of high-frequency details. This observation indicates that adapter scaling is not merely a structure-enhancement parameter, but a key control variable that governs the balance between shape consistency and texture preservation. To mitigate this hidden shape-texture trade-off, we propose Timestep-Conditioned Adapter Scaling (TCAS). TCAS adjusts the adapter strength according to the stages of the denoising process: strong adapter correction is applied in the middle geometry-refinement stage to enhance structural alignment, whereas conservative scales are used in the early and late stages to avoid disrupting global generation and texture synthesis. Experiments on multiple evaluation sets show that TCAS maintains shape fidelity close to aggressive uniform scaling while achieving better signal fidelity and more stable texture preservation. These results suggest that, in adapter-augmented multi-view diffusion pipelines, adapter strength should be treated as a stage-dependent inference control variable rather than a fixed hyperparameter. TCAS requires no retraining, introduces negligible computational overhead, and can be used as a plug-and-play strategy for geometry-controlled multi-view texture generation tasks.

Keywords: multi-view diffusion model; 3D texture generation; adapter scaling; timestep-conditioned scaling; shape-texture trade-off; geometric control; texture quality assessment

1 Introduction

Recent advances in single-image 3D generation have substantially improved the recovery of geometry, multi-view consistency, and sparse-view reconstruction [1-8]. As geometric generation models become increasingly mature, generating high-quality textures on an existing 3D shape has become a key factor affecting the realism and usability of 3D content. For an object whose geometry has already been determined, the final visual quality depends not only on whether the shape is complete, but also on whether the texture is faithful to the reference image, aligned with the 3D surface, and rich in local details.

Existing multi-view texture generation methods usually generate RGB images from multiple target views using 2D image diffusion models, and then map the generated images back to the 3D surface through projection or texture baking [9-14]. This route benefits from the generative capability of image diffusion models, but it still faces challenges in reference appearance alignment, geometry-texture consistency, and local texture detail preservation. Geometry-controlled multi-view diffusion methods represented by MVPainter introduce geometric conditions such as normals and depth maps to improve the alignment between generated textures and the underlying 3D surface, demonstrating the importance of explicit geometric control for 3D texture generation [9].

However, introducing geometric conditions is not only a question of whether a control signal should be used, but also a question of how strongly it should be applied. In adapter-augmented diffusion pipelines, geometric conditions are usually injected into the model through an adapter or control branch, and their influence is regulated by a scaling factor. In practice, this factor is often set empirically: a small scale may lead to insufficient geometric constraints, whereas a larger scale is usually expected to improve structural consistency.

In multi-view texture generation, however, stronger adapter scaling is not necessarily better. Texture generation requires not only contours and structures to be consistent with geometry, but also surface color variations, material details, and high-frequency patterns to be preserved. Overly strong geometric control may force the model to follow normal, depth, or edge conditions too aggressively, thereby suppressing the intrinsic texture synthesis capability of the base diffusion model. In this case, some structural metrics may still improve, while the visual result suffers from texture flattening, reduced color variation, and detail loss.

Therefore, adapter scaling in geometry-controlled multi-view diffusion deserves further analysis. Compared with prior work that mainly studies how to build texture generation pipelines, how to introduce geometric conditions, and how to improve overall generation quality, adapter strength is a fine-grained inference-time control problem. It directly affects the balance between shape consistency and texture detail, and is an easily overlooked factor in multi-view texture generation quality.

The main contributions of this paper are as follows:

(1) We systematically diagnose the shape-texture trade-off caused by adapter scaling in multi-view diffusion texture generation, showing that higher scales can improve structural metrics while simultaneously weakening color variation, flattening surfaces, and reducing high-frequency details.

(2) We propose Timestep-Conditioned Adapter Scaling (TCAS), which concentrates strong geometric correction in the middle denoising stage and uses conservative scales in the early and late stages, thereby achieving a more stable balance between geometric alignment and texture preservation.

(3) Through uniform scale sweeps, texture audits, analysis of 17 scheduling variants, and a 300-object validation, we demonstrate that adapter strength should be considered a stage-dependent inference control variable rather than a single fixed hyperparameter.

(4) We further provide a texture-loop analysis on the 300-object validation set, showing that TCAS substantially reduces texture flattening compared with full high-scale scaling: C3 outperforms s=2.50 on 87.7%, 95.0%, and 83.0% of objects in Laplacian variance, RGB standard deviation, and gradient magnitude, respectively, while the 95% confidence interval for the FG-SSIM difference crosses zero, indicating no significant difference in structural similarity.

2 Related Work

2.1 3D Texture Generation and Multi-view Diffusion

Single-image 3D generation is often decomposed into geometry generation and texture generation. In recent years, single-image-to-3D, multi-view diffusion, and sparse-view reconstruction methods have made notable progress in structural stability, shape completeness, and cross-category generalization [1-8]. Nevertheless, high-quality texture generation remains a key bottleneck for the realism and usability of 3D content. Existing texture generation methods usually rely on 2D diffusion models to synthesize RGB images from multiple views, and then project or bake these images onto the 3D surface [11-14]. While this strategy inherits the strong generative capability of image diffusion models, it still suffers from reference appearance misalignment, geometry-texture inconsistency, and insufficient local texture detail.

To address these issues, methods such as MVPainter formulate 3D texture generation as a geometry-controlled multi-view diffusion task [9]. MV-Adapter further shows that adapter-style multi-view control can enhance cross-view consistency without fully retraining the base model [10]. By injecting geometric information such as normals and depth maps into the diffusion model as conditional signals, the generated results better follow the 3D surface structure and remain consistent across views. Unlike prior methods that focus mainly on multi-view image generation or general 3D reconstruction, these works emphasize precise alignment between texture and geometry as well as local detail recovery, significantly improving 3D texture generation quality. However, they mainly answer how to introduce geometric conditions and how to construct a texture generation pipeline, while the influence of geometric control strength at inference time on shape and texture quality remains underexplored.

2.2 Adapter-augmented Diffusion Models and Geometric Control

Adapter modules have become an important mechanism for introducing additional conditions into diffusion models. ControlNet injects spatial conditions such as edges, depth maps, and normals into a pretrained diffusion model through a parallel branch [15]. IP-Adapter introduces image prompts through decoupled attention [16]. LoRA and its variants implement lightweight model adaptation through low-rank residual updates [17-19]. Although these methods differ in architecture, they generally include some form of scaling factor that controls the influence of the external condition or adapter residual on the base model.

In practice, the adapter scale is often treated as an empirical hyperparameter. A larger scale usually improves condition adherence, for example by better following normals, depth maps, or reference images. In texture generation tasks, however, overly strong conditional constraints may weaken the base diffusion model's ability to synthesize color, material, and high-frequency details. Adapter scaling is therefore not a simple matter of increasing the strength as much as possible; it involves a balance between geometric constraints and texture expression. This paper focuses on exactly this inference-time control problem: given a fixed adapter structure and fixed geometric conditions, how should the adapter strength be set or scheduled to preserve both shape consistency and texture detail?

2.3 Diffusion Guidance Scheduling and Timestep Control

The denoising process of diffusion models is highly stage-dependent. Early steps mainly determine global layout and coarse structure, middle steps refine shape and semantic structure, and late steps are more responsible for color, texture, and high-frequency details [20,21]. Based on this observation, many guidance scheduling strategies have been explored in both research and practice. For example, classifier-free guidance scales may be adjusted linearly, with cosine schedules, or dynamically to mitigate oversaturation, detail loss, or reduced diversity caused by overly strong guidance [22,23]. In addition, the development of latent diffusion, high-resolution diffusion, and Transformer-based diffusion models further shows the close relationship among generation quality, noise schedules, conditional injection, and network architecture [28-30,33-35].

These methods all exploit the staged nature of denoising to regulate conditional strength. However, different control signals act through different mechanisms. CFG scheduling adjusts the weight between conditional and unconditional predictions, mainly affecting the balance among semantic control, diversity, and visual fidelity. By contrast, the scale of a geometric adapter acts directly on intermediate features or residual branches, controlling how strongly geometric conditions intervene in the generation process. In multi-view texture generation, this intervention affects not only alignment with normals and depth maps, but also the base diffusion model's synthesis of color, material, and high-frequency texture. Therefore, timestep scheduling of adapter strength cannot be simply equated with CFG scheduling; it requires a dedicated analysis of the relationship between geometric control and texture generation.

3 Method

3.1 Task Definition and Base Pipeline

This paper builds on the MVPainter-style geometry-controlled multi-view texture generation framework [9] and studies how the inference-time strength of geometric condition injection affects the generated results. Given a single reference image and the corresponding 3D geometry, the base pipeline first generates RGB texture images from six target views and then obtains a textured mesh through projection or texture baking. We do not modify the base UNet, VAE, or pretrained adapter weights. Instead, the adapter scaling factor is treated as the key inference-time control variable.

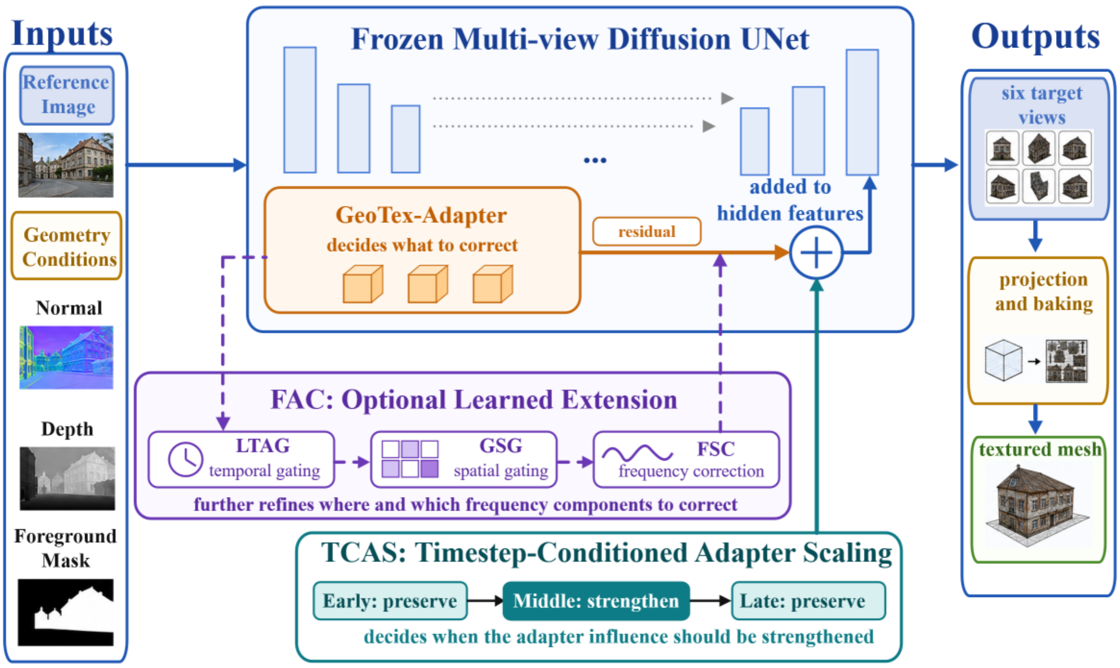

Figure 1. Method overview. GeoTex-Adapter generates geometric residuals and injects them into UNet features. TCAS dynamically adjusts the residual strength according to denoising stages to balance geometric consistency and texture fidelity. FAC is an optional extension module that further refines residual modulation along temporal, spatial, and frequency dimensions. The final multi-view outputs are projected and baked into a textured 3D mesh.

During multi-view diffusion, the 3D geometry is rendered into normal maps, depth maps, and foreground masks, denoted as:

$$G = \{G\_normal, G\_depth, G\_mask\}.$$

The geometric adapter maps these conditions to a residual correction term and injects it into the intermediate hidden states of the UNet. For the t-th denoising step, the adapter injection can be written as:

$$h'\_t = h\_t + s(p\_t) \cdot A\_phi(h\_t, G),$$

where h_t and h'_t denote the hidden states before and after injection, p_t denotes the sampling progress, and s(p_t) is the adapter scaling strength at the current denoising stage. A larger s usually enhances geometric condition adherence. However, if strong residuals are continuously applied in the late denoising stage, they may suppress the synthesis of color, material, and high-frequency details by the base diffusion model. This paper therefore studies how to schedule s during denoising to balance shape consistency and texture detail preservation.

3.2 Shape-Texture Diagnostic Metrics

The influence of adapter scaling on generated results cannot be evaluated by a single structural metric. We divide evaluation metrics into three groups. The first group consists of shape and structure metrics, including FG-SSIM, Edge-SSIM, and Crop-SSIM, which measure consistency with ground-truth renderings in the foreground, edge regions, and local foreground crops. The second group contains signal fidelity and perceptual similarity metrics, including PSNR and LPIPS, which reflect pixel-level error and differences in deep feature space. The third group consists of texture diagnostic metrics, including Laplacian variance, RGB standard deviation, and gradient magnitude, which are used to detect weakened color variation, surface smoothing, and high-frequency detail loss.

Taking Laplacian variance as an example, the foreground texture response is defined as:

$$M\_tex = Var(Lap(I\_out \odot G\_mask)),$$

where Lap denotes the Laplacian filter, G_mask denotes the foreground mask, and Var denotes spatial variance. A larger M_tex usually corresponds to richer local high-frequency variation. If structural metrics improve while Laplacian variance, RGB standard deviation, or gradient magnitude decreases, the improvement may mainly come from contour smoothing or boundary convergence rather than true texture quality improvement.

3.3 Timestep-Conditioned Adapter Scaling

The diffusion denoising process is stage-dependent: early steps mainly establish the global layout and coarse contours; middle steps refine geometric structure and cross-view alignment; late steps are more responsible for color, material, and local texture synthesis. Based on this observation, we propose Timestep-Conditioned Adapter Scaling (TCAS), which replaces a globally fixed adapter scale with a stage-dependent scale function.

To avoid ambiguity caused by different raw timestep directions in different schedulers, we use sampling progress p to represent the denoising process, where p = 0 denotes the start from noise and p = 1 denotes the stage close to the final image. TCAS divides denoising into early, middle, and late stages:

$$s(p) = \{ s\_e, 0 \le p < 1/3; \;\; s\_m, 1/3 \le p < 2/3; \;\; s\_l, 2/3 \le p \le 1 \}.$$

The final C3 configuration used in this paper is:

$$(s\_e, s\_m, s\_l) = (1.25, 2.50, 1.25).$$

This design follows three principles. First, the early stage uses a conservative scale to avoid disrupting global structure formation with strong geometric residuals. Second, the middle stage uses a higher scale to intensively strengthen geometric constraints from normals, depth maps, and foreground contours. Third, the late stage reduces the scale again to prevent geometric residuals from suppressing color variation, material detail, and high-frequency texture synthesis. TCAS introduces no new parameters and does not modify the base model or adapter weights; it only replaces the scaling factor during inference, so its computational overhead is negligible.

3.4 GeoTex-Adapter and FAC Adaptive Extension

GeoTex-Adapter, TCAS, and FAC correspond to three complementary dimensions of adapter control. GeoTex-Adapter determines what geometric residual correction should be injected. TCAS determines when the residual should be strengthened or weakened along the denoising-time dimension. FAC further modulates the residual in a fine-grained manner along spatial positions and frequency components.

GeoTex-Adapter is used as a fixed geometric residual backbone. It receives a 5-channel geometric input consisting of a 3-channel normal map, a 1-channel depth map, and a 1-channel foreground mask. A multi-scale geometric encoder produces a feature pyramid, and residual corrections are injected into multiple resolution levels of the UNet. TCAS acts on the global temporal scale of these residuals to control the influence of geometric conditions at different denoising stages without changing the adapter architecture.

To further examine whether stage-wise adapter control can be extended to learned adaptive correction, we also conduct experiments with Full Adaptive Correction (FAC). FAC contains three lightweight modules: LTAG learns timestep-dependent gating, GSG generates spatial gates from geometric features, and FSC performs frequency-selective correction on adapter residuals. Unlike fixed TCAS, FAC finely modulates adapter residuals in the temporal, spatial, and frequency domains. In the extension experiments, FAC uses TCAS as the base temporal schedule and further adds GSG and FSC for spatial and frequency correction.

It should be emphasized that FAC requires learning a small number of additional parameters. Therefore, TCAS, which requires no training, remains the main method in this paper. FAC is used only as a learned adaptive extension to show that stage-wise adapter control can be generalized to finer-grained adaptive correction.

4 Experiments

4.1 Experimental Setup

Experiments are conducted using an MVPainter-style geometry-controlled multi-view diffusion pipeline. Each object contains a single reference image and corresponding 3D geometry. During preprocessing, the 3D geometry is rendered into target-view normal maps, depth maps, foreground masks, and ground-truth RGB images. The model generates six target views at a time, and all comparisons with ground-truth renderings are performed under unified views, resolution, and foreground masks.

Except for the FAC extension experiment, all scaling experiments keep the object list, reference image, target views, random seed, base UNet, VAE, adapter weights, and number of denoising steps fixed; only the adapter scale or schedule is changed. The number of sampling steps is 50. C3 is selected only on a 24-object probe set, and no parameter search is performed on the 300-object validation set.

We use four evaluation sets: (1) a 50-object scale-sweep set for analyzing the overall trend of structural metrics under 13 uniform scales; (2) a 26-object texture audit set for comparing texture degradation under $s = 1.25$, $s = 2.25$, and $s = 2.50$; (3) a 24-object probe set for evaluating 17 uniform, layer-wise, and timestep scheduling variants and selecting C3; and (4) a 300-object validation set for comparing $s = 1.25$, $s = 2.50$, TCAS, and the FAC extension.

Evaluation metrics are divided into three categories: shape and structure metrics (FG-SSIM, Edge-SSIM, and Crop-SSIM), signal and perceptual metrics (PSNR and LPIPS), and texture diagnostic metrics (Laplacian variance, RGB standard deviation, and gradient magnitude). All metrics are first computed over six target views and then averaged over views and objects. For texture-related conclusions, we report means, object-level win rates, and significance tests to avoid judging generation quality based on a single structural score.

The FAC extension uses the same object list and evaluation protocol as the 300-object validation. TCAS is used as the main baseline, while LTAG, LTAG+GSG, and Full FAC are evaluated as ablation variants. Full FAC adds GSG and FSC on top of the TCAS temporal schedule. The base model and original adapter remain frozen, and only lightweight adaptive correction parameters are learned to verify the extensibility of the TCAS idea toward learned spatial and frequency correction.

4.2 Shape-Texture Trade-off of Adapter Scaling

Table 1 reports the results of a 13-scale uniform adapter sweep on 50 objects.

Table 1. Uniform adapter scale sweep results.

Scale	ΔFG-SSIM↑	POS	ΔCrop-SSIM↑	ΔPSNR↑
0.75	-0.013	20/50	+0.113	7.53
0.90	+0.024	30/50	+0.137	8.32
1.00	+0.047	34/50	+0.147	8.79
1.10	+0.070	36/50	+0.159	9.08
1.15	+0.071	36/50	+0.157	9.25
1.25	+0.088	37/50	+0.161	9.37
1.35	+0.100	39/50	+0.163	9.63
1.50	+0.113	40/50	+0.163	9.76
1.60	+0.120	42/50	+0.166	9.83
1.75	+0.127	43/50	+0.173	9.97
2.00	+0.134	44/50	+0.169	9.96

2.25	+0.142	43/50	+0.175	9.91
2.50	+0.145	44/50	+0.165	9.91

As the scale increases from s = 0.75 to s = 2.50, ΔFG-SSIM rises from -0.013 to +0.145, and the number of objects with positive improvement increases from 20/50 to 44/50. This indicates that stronger adapter residuals indeed improve the model's adherence to geometric conditions such as normals, depth maps, and foreground contours.

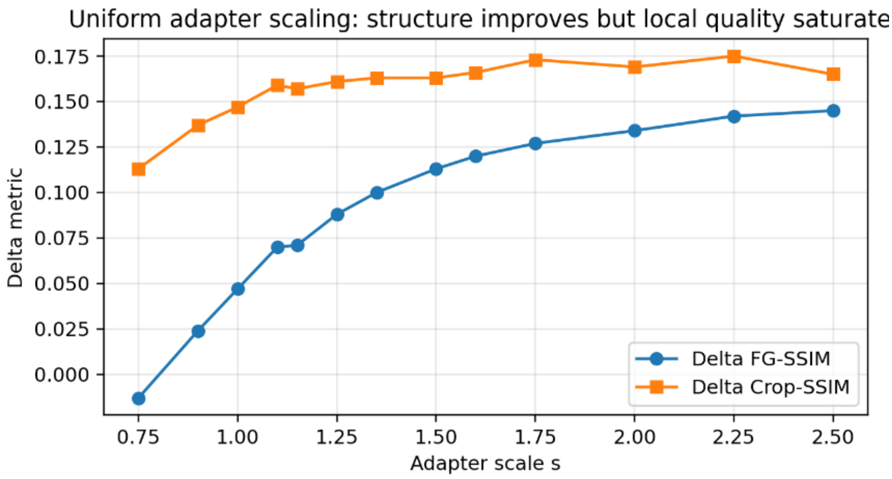

Figure 2. Structural metric changes under uniform adapter scaling. The curves show that FG-SSIM keeps increasing with adapter scale, while Crop-SSIM saturates and slightly decreases at high scales. Stronger geometric control therefore mainly improves structural alignment, but does not necessarily keep improving local foreground quality.

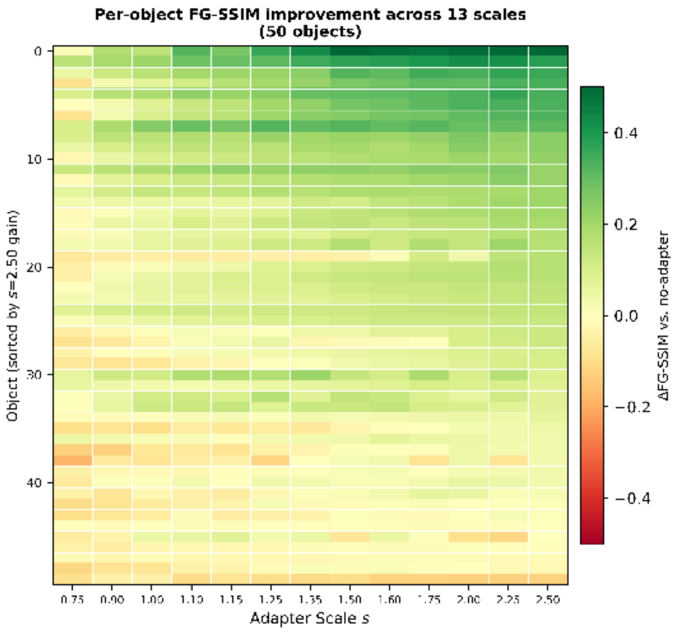

Figure 3. ΔFG-SSIM heatmap for 50 objects under 13 adapter scales. Most objects obtain higher structural gains as the scale increases, but there are clear object-level differences.

Overall, the FG-SSIM improvement is stable, but the marginal benefit becomes smaller in the high-scale region. For example, increasing the scale from s = 2.25 to s = 2.50 yields only a limited FG-SSIM gain, while Crop-SSIM already decreases. This suggests that the nature of the high-scale benefit may change: the gain comes more from contour smoothing and foreground alignment than from genuine restoration of local texture details.

4.3 Texture Audit: High Scale Causes Texture Flattening

To determine whether the FG-SSIM gain at high scales reflects a real improvement in visual quality, we further conduct a texture audit on 26 objects, comparing texture metrics and foreground shape changes for s = 2.50 relative to s = 1.25. Figure 4 shows qualitative results at different scales. The conservative scale s = 1.25 better preserves color variation and local texture, while s = 2.50 more easily produces uniform and smooth surfaces.

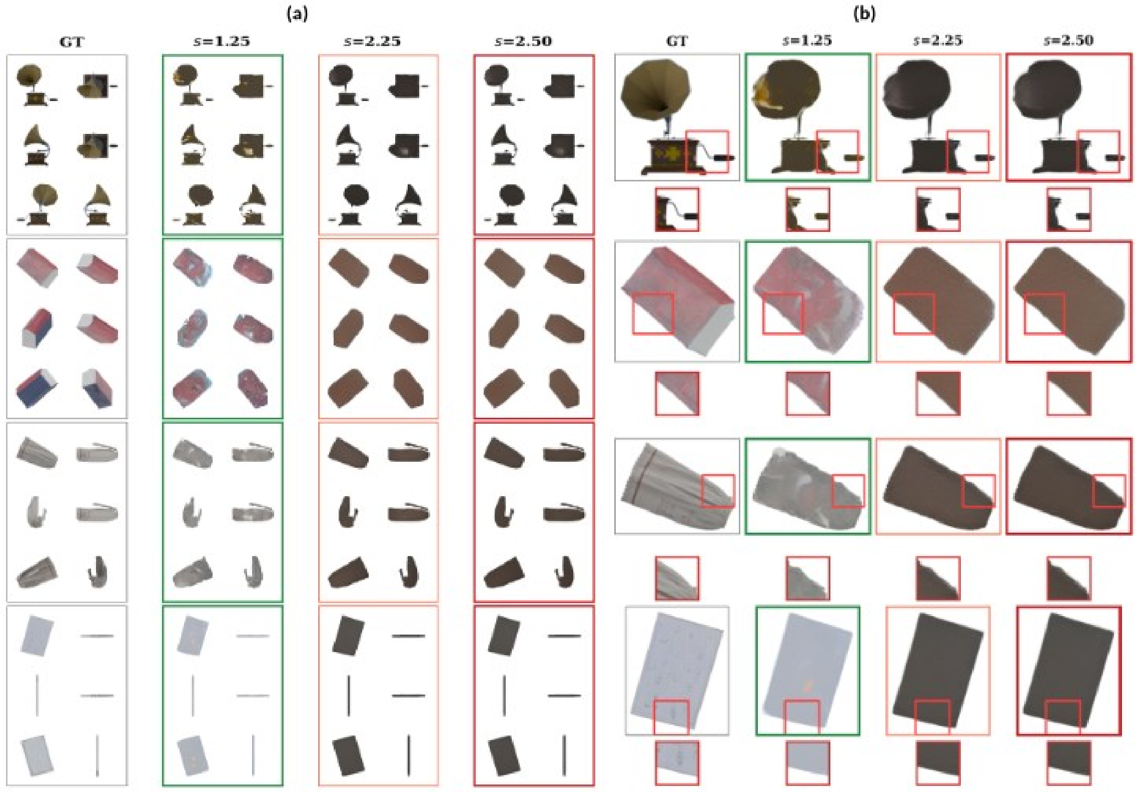

Figure 4. Texture degradation under different adapter scales. (a) Multi-view texture comparison. Each object is shown from six target views. Compared with conservative scaling s = 1.25, higher scales s = 2.25 and s = 2.50 show more obvious color convergence and surface smoothing in multiple views. (b) Local zoom-in comparison. Red boxes mark local texture regions; green borders denote conservative scaling s = 1.25, and red borders denote aggressive scaling s = 2.50. As the adapter scale increases, local color variation, material details, and high-frequency texture gradually weaken, and the surface becomes more uniform.

The results show that 24 out of 26 objects (92%) exhibit a pattern of “texture loss without true shape gain.” Only two objects show comparable texture quality, and no object shows a clear texture improvement from the high scale. Here, “true shape gain” is not equivalent to an increase in FG-SSIM. It refers to whether the high scale brings substantive improvement in effective foreground area, contour structure, or geometric detail compared with the conservative scale. If the metric improvement mainly comes from smoothed foreground boundaries or averaged contour errors without visible geometric improvement, it is categorized as having no true shape gain. Therefore, although a high adapter scale can improve structural metrics, such improvement does not necessarily indicate better real shape or texture quality.

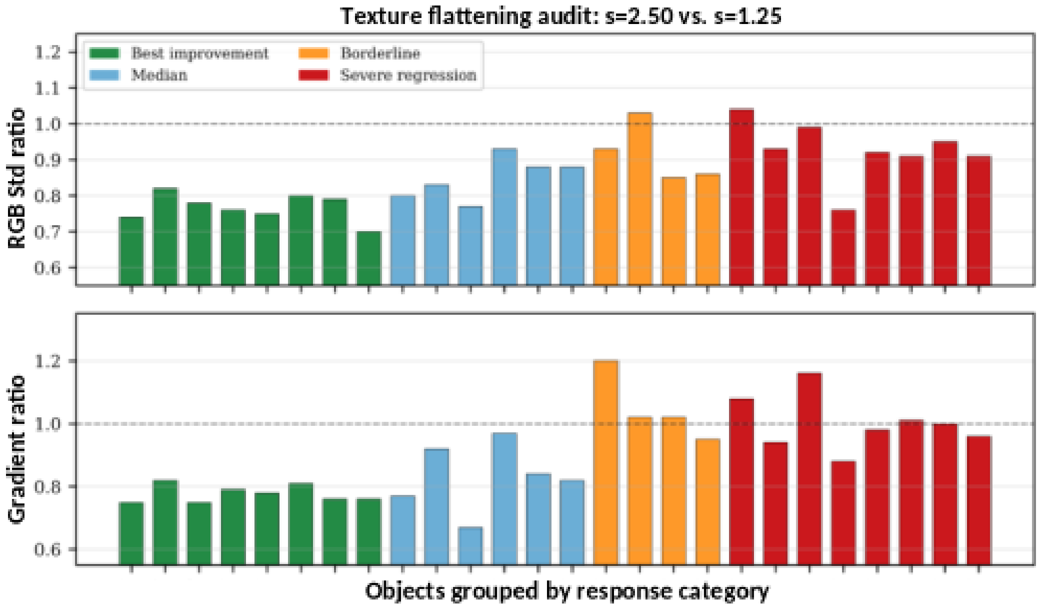

Figure 5. Texture flattening at s = 2.50. Top: RGB standard deviation ratio on 26 objects (s = 2.50 / s = 1.25). Bottom: gradient magnitude ratio. Values below 1.0 indicate texture loss. Even the four best-improvement objects (green) show 24% to 30% texture loss despite positive FG-SSIM gains.

For multi-view texture generation, overly strong geometric control can suppress the color and detail synthesis capability of the base diffusion model during late denoising steps, making surfaces more uniform and smoother. Therefore, scale selection should not rely only on standard metrics such as FG-SSIM or PSNR, but should also incorporate texture-sensitive diagnostics.

4.4 Timestep-Conditioned Scaling: Shape-Texture Balance of C3

Based on the texture audit, we no longer pursue higher adapter scales throughout the entire denoising process. Instead, we further explore how different scaling strategies balance shape gain and texture loss. We construct 17 adapter scaling variants and evaluate them on the 24-object probe set. These variants include three categories: uniform scaling baselines, layer-wise scaling strategies, and timestep-conditioned scaling strategies. Uniform scaling tests the effect of globally strengthening adapter influence; layer-wise scaling examines the strength of geometric injection at different UNet levels; and timestep scaling verifies the sensitivity of different denoising stages to geometric control strength.

All variants are evaluated using ΔFG-SSIM for shape gain and ΔLap Var for texture detail change. Importantly, the objective of this probe experiment is not to find the single highest structural score, but to identify a stable compromise strategy that avoids the trap of "higher structural metrics but flattened texture."

Table 2. Shape-texture trade-off on the 24-object probe set. ΔLap Var closer to 0 indicates smaller texture loss.

Variant	Category	ΔFG-SSIM↑	ΔLap Var → 0	Assessment
C4	Late-stage high scale	0.105	-0.0120	Trap
C2	Timestep scheduling	0.104	-0.0110	Trap
B2	Layer-wise scheduling	0.090	-0.0135	Trap
A_s2.00	Uniform scaling	0.087	-0.0062	Trap
C3 (TCAS)	Mid-stage high scale	0.084	-0.0006	Promising
A_s2.25	Uniform scaling	0.079	-0.0096	Trap
A_s2.50	Uniform scaling	0.078	-0.0108	Trap
B6	Layer-wise scheduling	0.000	-0.0027	OK

Table 2 summarizes representative scheduling variants on the 24-object probe set. C4 and other late-stage high-scale strategies obtain higher ΔFG-SSIM, but they are accompanied by obvious decreases in Laplacian variance, indicating that their structural gains come at the cost of texture loss. Uniform high-scale strategies show a similar problem. For example, s = 2.50 achieves ΔFG-SSIM of +0.078, but ΔLap Var decreases to -0.0108. In contrast, C3 achieves ΔFG-SSIM of +0.084. Although it is slightly lower than some aggressive variants in structure gain, its Laplacian variance loss is only -0.0006, close to zero texture loss. Therefore, C3 is the only variant labeled Promising and is used as the TCAS configuration in subsequent large-scale validation.

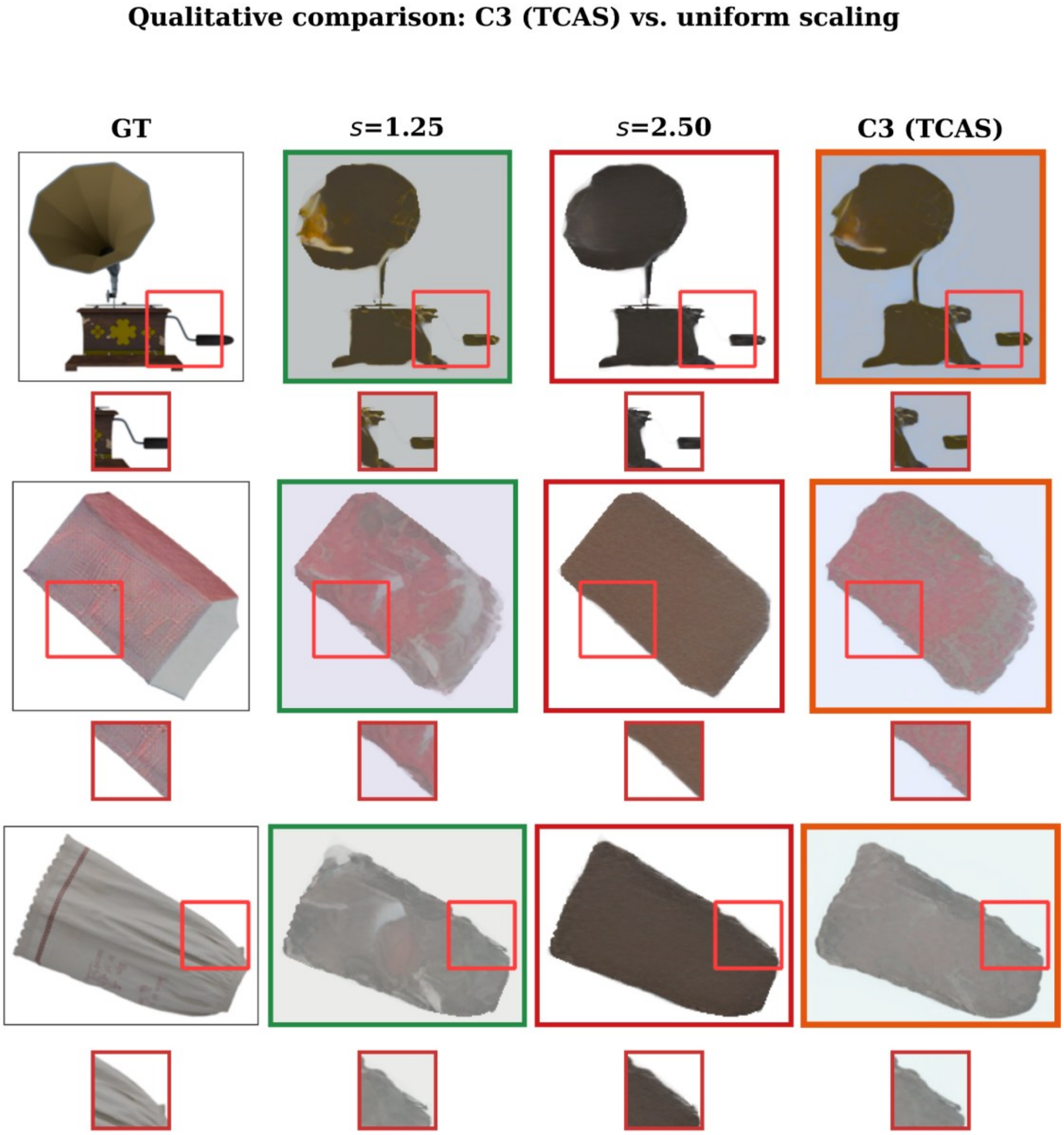

Figure 6. Qualitative comparison between C3 and uniform adapter scaling. Compared with conservative scaling s = 1.25, aggressive scaling s = 2.50 provides stronger structural constraints, but it easily causes color convergence, surface flattening, and local texture loss. C3 applies strong geometric correction in the middle stage and reduces adapter intervention in the late stage. It preserves a certain degree of shape alignment while reducing the obvious texture flattening observed in s = 2.50. Red boxes indicate local zoom-in regions.

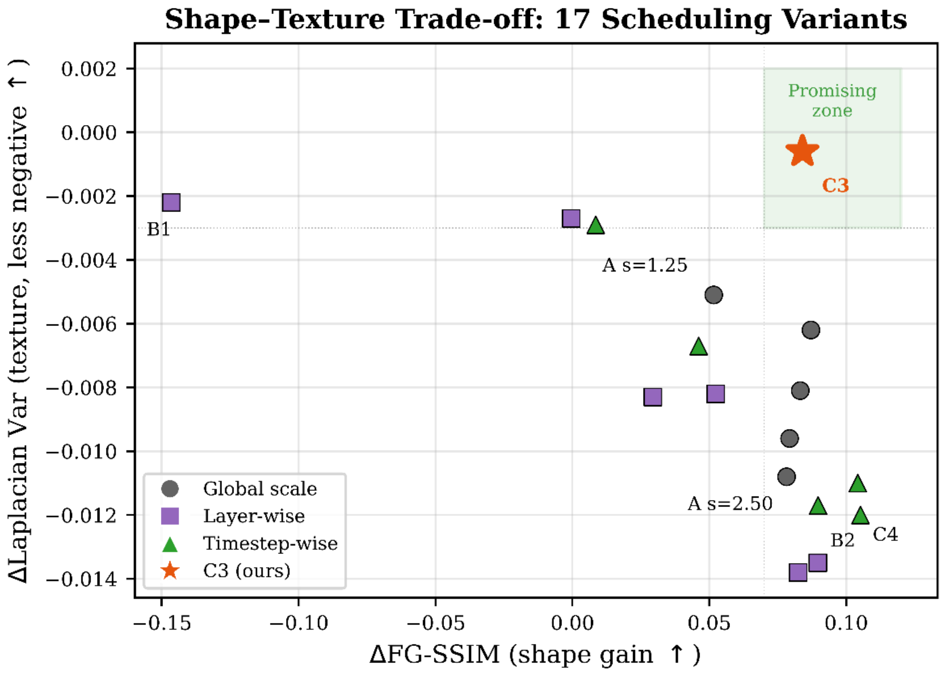

Figure 7. Shape-texture Pareto trade-off among 17 variants. The horizontal axis is shape gain, where farther right is better. The vertical axis is texture loss, where lower is better; therefore, the bottom-right region represents a more desirable compromise. C3 (star) lies in the Promising region, with high shape gain and nearly zero texture loss. Most high-shape-gain variants fall into the Trap region, where structural scores are obtained at the cost of large texture loss. Variants such as B6 have smaller texture degradation but almost no shape gain.

The specific configuration of C3 uses the conservative scale s = 1.25 in the early and late stages, and the strong scale s = 2.50 in the middle geometry-refinement stage, i.e., C3 = (1.25, 2.50, 1.25). Its motivation follows the staged role of diffusion denoising. In the early stage, latent variables are still highly noisy, and strong geometric correction can interfere with global structure formation. In the middle stage, the signal becomes more stable, making it suitable to strengthen geometric constraints such as normals, depth maps, and foreground contours. In the late stage, generation mainly focuses on color, material, and fine texture synthesis, so adapter intervention should be reduced to preserve the texture generation space of the base diffusion model.

Figure 7 further shows the distribution of the 17 variants in the shape-texture space. Most variants with high shape gain are accompanied by large texture loss and fall into the Trap region, whereas C3 maintains a relatively high shape gain while keeping texture loss near zero.

Thus, the value of C3 is not that it maximizes every metric, but that it avoids the trap of “higher structural scores but flattened texture.” It preserves the advantage of using the adapter for geometry correction in the middle stage, while avoiding excessive late-stage interference with texture synthesis, resulting in a more stable shape-texture compromise.

Mechanistically, the advantage of C3 is not simply the result of hyperparameter tuning; it comes from distinguishing the semantic roles of different denoising stages. Early latent variables are under high noise, so strong adapter correction is unlikely to produce stable geometric gains and may disturb global structure formation. In the middle stage, signals become clearer, and geometric conditions such as normals, depth maps, and foreground contours are most effective. In the late stage, color, material, and fine texture are synthesized; continuously applying strong adapter residuals at this point tends to suppress the texture generation capability of the base diffusion model.

This explanation is consistent with the scheduling-variant results. Early high-scale or early-and-late high-scale variants provide limited structural gain, indicating that early strong intervention is unstable. The late-stage high-scale configuration C4 obtains a high structural score but shows a significant decrease in Laplacian variance, making it a typical Trap variant where structural metrics improve while texture is flattened. Aggressive C3 further increases the middle-stage strength but reduces signal fidelity, suggesting that even middle-stage correction has diminishing returns and can lead to over-constraint.

Table 3. Comparison of representative variants under extended texture metrics. This table verifies that the advantage of C3 is not due to a single Laplacian variance metric.

Variant	ΔLap Var	ΔRGB Std	ΔGrad	ΔHF Energy
C3	-0.001	-0.023	+0.142	+0.058

C4	-0.012	-0.032	+0.086	+0.062
A_s2.50	-0.011	-0.044	+0.054	+0.078
A_s2.00	-0.006	-0.040	+0.100	+0.070
A_s1.25	-0.005	-0.032	+0.104	+0.053

C3 has nearly zero loss in Laplacian variance and remains relatively stable in RGB standard deviation, gradient magnitude, and high-frequency energy. Compared with C4 and uniform high-scale strategies, C3 shows weaker texture degradation, indicating that its shape-texture balance is not an artifact of a single metric.

4.5 Large-scale Validation

Finally, we compare conservative uniform scaling (s = 1.25), aggressive uniform scaling (s = 2.50), and fixed C3 (TCAS) on 300 objects. C3 is not re-searched on the 300-object set; it directly uses the timestep schedule selected on the 24-object probe set to evaluate the transferability of the strategy to a larger object set.

Table 4. Large-scale validation on 300 objects. Values are absolute metrics of generated outputs with respect to ground truth.

Method	FG-SSIM↑	Edge-SSIM↑	PSNR↑	FG-LPIPS↓
s = 1.25	0.430	0.513	17.95	0.174
s = 2.50	0.473	0.559	17.90	0.172
C3 (TCAS)	0.476	0.529	18.86	0.171

As shown in Table 4, conservative uniform scaling s = 1.25 has lower FG-SSIM and Edge-SSIM, indicating insufficient geometric constraints. Aggressive uniform scaling s = 2.50 obtains the highest Edge-SSIM, suggesting that continuously strengthening adapter residuals helps edge and contour alignment. However, its PSNR is lower than that of C3, indicating that overly strong geometric intervention reduces overall signal fidelity. In contrast, C3 achieves an FG-SSIM of 0.476, which is comparable to s = 2.50, while increasing PSNR to 18.86 and slightly reducing FG-LPIPS. These results show that timestep-conditioned scaling preserves the main shape fidelity while improving signal quality.

PSNR mainly reflects global pixel error and should not be directly equated with texture detail quality. To further verify texture preservation, we compute Laplacian variance, RGB standard deviation, and gradient magnitude on the same 300 objects to measure foreground high-frequency response, color variation amplitude, and local gradient strength, respectively. All results use the same object list, random seed, 50 denoising steps, and adapter weights to ensure comparability.

Table 5. Large-scale texture-loop validation on 300 objects. Higher texture metrics indicate better preservation of foreground color variation and high-frequency details.

Method	Lap Var↑	RGB Std↑	Grad Mag↑
s = 1.25	0.0151	0.0408	0.3258
s = 2.50	0.0095	0.0251	0.2616
C3 (TCAS)	0.0129	0.0377	0.2919

Table 5 shows that although s = 2.50 achieves the highest Edge-SSIM, it has the lowest values across all three texture metrics, indicating that full high-scale scaling weakens color variation and high-frequency details. C3 has FG-SSIM comparable to s = 2.50 while substantially improving Laplacian variance, RGB standard deviation, and gradient magnitude, demonstrating a better balance between structural alignment and texture preservation.

To further quantify texture loss and object-level stability, we jointly analyze texture retention ratios relative to s = 1.25 and object-level statistics of C3 relative to s = 2.50. The former measures the degree of texture loss relative to conservative scaling, and the latter examines whether the advantage of C3 consistently appears across most objects.

Table 6. Texture retention ratios and object-level stability of C3. Texture ratios are measured relative to s = 1.25; object-level win rates and confidence intervals for C3 over s = 2.50 are reported in the text.

Method	Lap Var Ratio↑	RGB Std Ratio↑	Grad Mag Ratio↑
s = 1.25	1.000	1.000	1.000
s = 2.50	0.678	0.643	0.807

C3 (TCAS)	0.835	0.970	0.883

As shown in Table 6, compared with conservative scaling s = 1.25, aggressive scaling s = 2.50 retains only 67.8% of Laplacian variance, 64.3% of RGB standard deviation, and 80.7% of gradient magnitude, indicating that full high-scale scaling substantially weakens foreground color variation and high-frequency details. In contrast, C3 retains 83.5%, 97.0%, and 88.3% of these metrics, respectively, and clearly outperforms s = 2.50 across all three texture measures. This shows that TCAS still incurs some texture reduction, but it substantially alleviates texture flattening compared with full high-scale scaling.

Object-level statistics further show that C3 outperforms s = 2.50 on 87.7%, 95.0%, and 83.0% of objects in Laplacian variance, RGB standard deviation, and gradient magnitude, respectively. The corresponding Bootstrap 95% confidence intervals do not cross zero, indicating statistically stable texture-preservation advantages. C3 also outperforms s = 2.50 on 74.3% of objects in PSNR. By contrast, the confidence interval for FG-SSIM crosses zero, indicating no significant difference in foreground structural similarity between C3 and s = 2.50. Therefore, TCAS does not noticeably sacrifice structural alignment, while significantly reducing the texture degradation caused by full high-scale scaling.

In summary, the 300-object experiments demonstrate that adapter scaling affects both structural alignment and texture detail preservation. Full high-scale scaling strengthens edge constraints but is more likely to cause texture flattening. TCAS strengthens geometric correction in the middle stage and reduces adapter intervention in the late stage, thereby maintaining structural alignment while substantially reducing texture loss. It should be noted that C3 still has some texture decrease relative to s = 1.25; therefore, we do not claim that TCAS completely eliminates texture loss, but rather that it significantly reduces texture degradation compared with full high-scale scaling.

4.6 FAC Adaptive Correction Extension

TCAS uses a fixed three-stage scale, which has the advantages of requiring no training, being simple to implement, and introducing low computational overhead. However, its residual scaling is still uniform across spatial positions and frequency components, lacking fine-grained adaptivity. To verify whether stage-wise adapter control can be further extended, we evaluate FAC adaptive correction on the 300-object validation set. FAC does not modify the base multi-view diffusion model, VAE, or original GeoTex-Adapter weights. Instead, it introduces lightweight modules on top of the TCAS temporal schedule: LTAG for timestep gating, GSG for spatial gating, and FSC for frequency-selective correction. Because FAC learns a small number of additional parameters, it is treated only as an extension experiment, while the main method remains the training-free TCAS.

Table 7 reports the controlled ablation results. The TCAS baseline already achieves strong performance. LTAG only performs worse than TCAS on all metrics, suggesting that learned temporal gating alone cannot stably replace the fixed three-stage schedule. LTAG+GSG improves SSIM to 0.8794, but its PSNR, FG-PSNR, and FG-SSIM remain lower than TCAS, showing that spatial gating alone is insufficient to improve foreground quality. Full FAC, which adds spatial gating and frequency-selective correction on top of TCAS, achieves the best overall performance: PSNR increases from 18.86 to 19.28, SSIM remains comparable at 0.8793, FG-PSNR increases from 11.66 to 11.77, and FG-SSIM increases from 0.4755 to 0.4830.

Table 7. Ablation results of the FAC adaptive correction extension on the 300-object validation set. Full FAC adds spatial gating and frequency-selective correction on top of the TCAS temporal schedule.

Variant	PSNR↑	SSIM↑	FG-PSNR↑	FG-SSIM↑
TCAS (baseline)	18.86	0.8737	11.66	0.4755
LTAG only	17.87	0.8732	10.47	0.4635
LTAG + GSG	18.09	0.8794	10.43	0.4640
Full FAC	19.28	0.8793	11.77	0.4830

Object-level statistics further show that Full FAC outperforms TCAS on 81.3% and 72.0% of objects in PSNR and FG-SSIM, respectively, and paired significance tests for both metrics are significant ($p < 0.0001$). This indicates that the gains of FAC are stable across most objects rather than being driven by a small number of samples.

Therefore, FAC does not change the main conclusion of this paper. TCAS remains the main method because it is training-free and plug-and-play. FAC serves as an extension experiment, showing that stage-wise adapter control can be further generalized to lightweight learned spatial and frequency adaptive correction.

5 Limitations

This paper mainly focuses on inference-time adapter scaling and does not retrain the base multi-view diffusion model or the adapter. Therefore, TCAS can alleviate texture flattening under high scales, but it cannot fundamentally solve the limitations of the base model in invisible-region completion, complex material synthesis, and cross-view detail consistency.

Second, Laplacian variance, RGB standard deviation, and gradient magnitude are proxy metrics for texture degradation. They are effective for detecting high-frequency loss and surface smoothing, but they cannot fully replace human perceptual evaluation. For objects with strong reflection, transparency, thin structures, or highly stylized textures, discrepancies may exist between these metrics and subjective quality.

In addition, the current TCAS uses fixed stage boundaries and a fixed scale combination, C3 = (1.25, 2.50, 1.25). This setting is simple and stable, but it has not yet been adapted to object complexity, material type, or input reference image quality. Future work could study content-adaptive stage boundaries and scale selection strategies.

Finally, although the FAC extension demonstrates the potential of learned spatial and frequency correction, it requires additional training of a small number of parameters and therefore cannot replace the training-free property of TCAS. We treat FAC as an extension validation rather than the main method.

6 Conclusion

This paper presents a systematic experimental analysis of adapter scaling in multi-view diffusion texture generation. The experiments show that higher adapter scales continuously improve structural metrics such as FG-SSIM, but such improvement can mask texture quality degradation. In the 26-object texture audit, 92% of objects show texture flattening and high-frequency detail loss under $s = 2.50$.

To address this shape-texture trade-off, we propose Timestep-Conditioned Adapter Scaling (TCAS). The best configuration, C3, applies strong adapter correction in the middle denoising stage and uses conservative scales in the early and late stages. The 24-object probe set shows that C3 maintains relatively high structural gain while substantially reducing Laplacian variance loss. The 300-object validation further shows that C3 achieves FG-SSIM comparable to aggressive uniform scaling $s = 2.50$ while improving PSNR by 0.96 dB.

Therefore, the core contribution of this paper is not to claim that the generated results are visually close to the ground truth in all cases, but to reveal and mitigate a hidden trade-off in adapter scaling. This conclusion suggests that, in geometry-controlled multi-view diffusion models, adapter strength should be treated as a stage-dependent inference control variable rather than a single fixed hyperparameter.

The FAC extension further demonstrates that the stage-wise adapter control idea can be used not only as a fixed inference schedule, but also generalized to lightweight learned spatial and frequency adaptive correction, providing a direction for future content-adaptive texture generation.

References

[1] LIU R, WU R, VAN HOORICK B, et al. Zero-1-to-3: Zero-shot one image to 3D object[C]//Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.

[2] SHI R, CHEN H, ZHANG Z, et al. Zero123++: A single image to consistent multi-view diffusion base model[EB/OL]. arXiv:2310.15110, 2023.

[3] LIU Y, LIN C, ZENG Z, et al. SyncDreamer: Generating multiview-consistent images from a single-view image[EB/OL]. arXiv:2309.03453, 2023.

[4] LONG X, GUO Y C, LIN C, et al. Wonder3D: Single image to 3D using cross-domain diffusion[EB/OL]. arXiv:2310.15008, 2023.

[5] SHI Y, WANG P, YE J, et al. MVDream: Multi-view diffusion for 3D generation[C]//International Conference on Learning Representations. 2024.

[6] WANG P, et al. ImageDream: Image-prompt multi-view diffusion for 3D generation[EB/OL]. arXiv:2312.02201, 2023.

[7] XU J, et al. InstantMesh: Efficient 3D mesh generation from a single image with sparse-view large reconstruction models[C]//Advances in Neural Information Processing Systems. 2024.

[8] KE Q, et al. Era3D: High-resolution multiview diffusion using efficient row-wise attention[EB/OL]. arXiv:2405.11616, 2024.

[9] SHAO M, XIONG F, SUN Z, XU M. MVPainter: Accurate and detailed 3D texture generation via multi-view diffusion with geometric control[EB/OL]. arXiv:2505.12635, 2025.
[10] HUANG Z, et al. MV-Adapter: Multi-view consistent image generation made easy[EB/OL]. arXiv:2412.03632, 2024.
[11] CHEN D Z, et al. Text2Tex: Text-driven texture synthesis via diffusion models[EB/OL]. arXiv preprint, 2023.
[12] RICHARDSON E, et al. TEXTure: Semantic texture transfer using diffusion models[EB/OL]. arXiv preprint, 2023.
[13] POOLE B, JAIN A, BARRON J T, MILDENHALL B. DreamFusion: Text-to-3D using 2D diffusion[C]//International Conference on Learning Representations. 2023.
[14] CHEN R, CHEN Y, JIAO N, JIA K. Fantasia3D: Disentangling geometry and appearance for high-quality text-to-3D content creation[C]//Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.
[15] ZHANG L, RAO A, AGRAWALA M. Adding conditional control to text-to-image diffusion models[C]//Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.
[16] YE H, ZHANG J, LIU S, HAN X, YANG W. IP-Adapter: Text compatible image prompt adapter for text-to-image diffusion models[EB/OL]. arXiv:2308.06721, 2023.
[17] HU E J, SHEN Y, WALLIS P, et al. LoRA: Low-rank adaptation of large language models[C]//International Conference on Learning Representations. 2022.
[18] LIU S Y, WANG C Y, YIN H, et al. DoRA: Weight-decomposed low-rank adaptation[EB/OL]. arXiv:2402.09353, 2024.
[19] ZHANG Q, ZUO S, LIANG C, et al. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning[C]//International Conference on Learning Representations. 2023.
[20] HO J, JAIN A, ABBEEL P. Denoising diffusion probabilistic models[C]//Advances in Neural Information Processing Systems. 2020.
[21] SONG J, MENG C, ERMON S. Denoising diffusion implicit models[C]//International Conference on Learning Representations. 2021.
[22] HO J, SALIMANS T. Classifier-free diffusion guidance[C]//NeurIPS Workshop. 2022.
[23] CASTILLO A, LEGENSTEIN R, et al. Adaptive guidance: Training-free acceleration of conditional diffusion models[EB/OL]. arXiv:2312.12487, 2023.
[24] WANG Z, BOVIK A C, SHEIKH H R, SIMONCELLI E P. Image quality assessment: From error visibility to structural similarity[J]. IEEE Transactions on Image Processing, 2004, 13(4): 600-612.
[25] ZHANG R, ISOLA P, EFROS A A, SHECHTMAN E, WANG O. The unreasonable effectiveness of deep features as a perceptual metric[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2018.
[26] RADFORD A, KIM J W, HALLACY C, et al. Learning transferable visual models from natural language supervision[C]//International Conference on Machine Learning. 2021.
[27] OQUAB M, DARCET T, MOUTAKANNI T, et al. DINOv2: Learning robust visual features without supervision[J]. Transactions on Machine Learning Research, 2024.
[28] ROMBACH R, BLATTMANN A, LORENZ D, ESSER P, OMMER B. High-resolution image synthesis with latent diffusion models[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022.
[29] ESSER P, KULAL S, BLATTMANN A, et al. Scaling rectified flow transformers for high-resolution image synthesis[C]//International Conference on Machine Learning. 2024.
[30] BLACK FOREST LABS. FLUX: Flow matching for image generation[EB/OL]. 2024.
[31] LOSHCHILOV I, HUTTER F. Decoupled weight decay regularization[C]//International Conference on Learning Representations. 2019.
[32] CHERTI M, BEAUMONT R, WIGHTMAN R, et al. Reproducible scaling laws for contrastive language-image learning[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023.
[33] NICHOL A, DHARIWAL P, RAMESH A, et al. GLIDE: Towards photorealistic image generation and editing with text-guided diffusion models[C]//International Conference on Machine Learning. 2022.
[34] SAHARIA C, CHAN W, SAXENA S, et al. Photorealistic text-to-image diffusion models with deep language understanding[C]//Advances in Neural Information Processing Systems. 2022.
[35] PEEBLES W, XIE S. Scalable diffusion models with transformers[C]//Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.